

CoolDesk

AC Service Management Portal

Design Requirements Document

v2.0

Audience: Designers · Engineers · QA

Derived from: UI/UX Specification v1.0 · User Flows v1.1 · Schema v2.2

Incorporates: UI/UX Recommendations Review (resolved decisions, critical fixes, role-based improvements)

Change Summary — v2.0

This revision integrates the UI/UX Recommendations review into the canonical Design Requirements. All previously open questions in v1.0 §7 are resolved here as binding requirements; all critical UX fixes and role-based improvements identified in the review are now first-class component, pattern, or system requirements.

Resolved (was §7 of v1.0)

- OQ-01 — Amber Alert KPI zero state: green zero-state with checkmark and "No revisits pending" label.
- OQ-02 — Pending Schedule Days Waiting threshold: 2-day amber / 4-day red (installation), 1-day amber / 2-day red (complaint); both are configurable.
- OQ-03 — Staff rollback from needs_revisit: revisit row preserved as historical record (60s technician undo deletes it; staff rollback does not).
- OQ-04 — Dealer VCID conflict visibility: "New Customer / Existing Customer" binary only; no further detail surfaced.
- OQ-05 — Auto-rejection trigger: service-layer check on every status transition; system_event type = cancellation_auto_rejected.
- OQ-06 — Offline banner: not user-dismissible; pinned until connectivity is restored.
- OQ-07 — Withdrawal reason: 10-character minimum enforced (consistent with all other reason textareas).
- OQ-08 — Optional review sheet dismissal: back-navigation and tap-outside both count as Skip in optional mode.
- OQ-09 — Dealer analytics tab: visible to both Owner and Office Staff.
- OQ-10 — Brand swatch fallback: neutral grey (#9E9E9E) when colour_hex is null.
- Icon library: Lucide Icons (system-wide).
- Animation tokens: anim-instant (100ms), anim-standard (250ms), anim-countdown (1000ms).

Newly required (was "recommended" in review)

- Status chips carry icons in addition to colour and label (WCAG 1.4.1).
- OS-02 Job Creation auto-saves draft state across the 4 steps.
- Undo chip is a global sticky overlay anchored to the app shell, not a per-screen element.
- Blocking modals visually advertise themselves ("Action Required" badge, no close affordance, shake-on-dismiss).
- Amber Alert Panel default sort: CHRONIC first, then by oldest last-visit date.
- Payment Form (TC-03) requires a one-screen confirmation before final submit.

- WebSocket disconnect produces an explicit "Live updates paused" indicator on the desktop portal.
- Filter state persists in URL query parameters across all job list surfaces.
- Notification Centre supports grouping, filter chips, and pagination at 50 items.
- Dealer portal exposes a simplified progress stepper and KPI row.
- VCID phone-lookup UI shows explicit confidence labels ("Best match" / "Possible match").
- Toasts on the Technician PWA appear at the top of the screen.
- HTTP 409 conflict resolution is a bottom sheet with a 3-way choice.
- Touch targets on the PWA enforce a 48px minimum height (WCAG 2.5.5).
- Primary action button respects iOS safe-area-inset-bottom.

New requirements introduced

- Section 1.5 — Icon System (Lucide assignments per status group, navigation, and timeline).
- Section 1.6 — Animation Tokens.
- Section 8 — Navigation Structure (per role).
- Section 9 — Workflow Optimisations (batch scheduling, route optimisation, smart pre-fill, urgency escalation, pre-chronic warning, command palette).
- New colour tokens: color-status-new, color-status-revisit (re-keyed to #EF6C00), color-surface-info, color-type-installation, color-type-complaint, color-source-direct, color-source-dealer, color-brand-fallback, color-rating-filled, color-rating-empty.

1. Foundations

1.1 Colour System

All values are design tokens. Tokens marked NEW were introduced in v2.0 to remove value-collisions and to give Job Type, Source, brand-fallback, and rating-state surfaces their own semantic identity.

Token Name	Hex	Semantic Role	Usage
color-primary-900	#1E3A5F	Brand / Primary dark	Section headers, table headers, primary text emphasis
color-primary-600	#2E75B6	Brand / Accent blue	Links, active states, borders, H2 underlines
color-primary-100	#D5E8F0	Brand / Surface light	Table alternate rows, structural surfaces only
color-surface-info (NEW)	#D5E8F0	Info banner surface	Replaces color-primary-100 for info banners and note boxes; allows independent evolution from structural use
color-status-pending	#9E9E9E	Status: Pending Schedule	pending_schedule chip
color-status-new (NEW)	#9E9E9E	Status: New Complaint	new (complaint) chip — split from color-status-pending so the two can evolve independently
color-status-scheduled	#1565C0	Status: Scheduled	scheduled chip
color-status-assigned	#303F9F	Status: Assigned	assigned chip (Indigo)
color-status-acknowledged	#00695C	Status: Acknowledged	acknowledged chip (Teal)
color-status-transit	#00838F	Status: In Transit	in_transit chip (Cyan)
color-status-inprocess	#F57F17	Status: In Process	in_process chip, amber alert border
color-status-completed	#2E7D32	Status: Completed / Resolved	completed, resolved, resolved_on_revisit chips
color-status-revisit (CHANGED)	#EF6C00	Status: Needs Revisit	needs_revisit chip + border. Re-keyed from #F57F17 to remove value-collision with color-status-inprocess
color-status-cancel-req	#E53935	Status: Cancellation Requested	cancellation_requested chip

Token Name	Hex	Semantic Role	Usage
color-status-cancelled	#4E342E	Status: Cancelled (terminal)	cancelled chip
color-feedback-error	#C00000	Error / Destructive	Field errors, delete actions, CHRONIC tag
color-feedback-warning-bg	#FFF2CC	Warning background	Amber alert panels, edge case boxes
color-feedback-warning-border	#FFC000	Warning border	Left border on warning notes
color-feedback-success-bg	#E2EFDA	Success / positive	Acceptance criteria boxes, resolved states, Amber Alert KPI zero state
color-surface-alt	#F2F2F2	Table alternate row	Zebra striping on tables
color-text-primary	#404040	Primary body text	All body copy
color-text-muted	#808080	Muted / secondary	Relative timestamps, system event labels
color-type-installation (NEW)	#37474F	Job Type: Installation	Installation chips on lists, cards, detail headers
color-type-complaint (NEW)	#BF6E2A	Job Type: Complaint	Complaint chips on lists, cards, detail headers
color-source-direct (NEW)	#1565C0	Job Source: Direct	Direct source chips on lists, cards, detail headers
color-source-dealer (NEW)	#5C6BC0	Job Source: Via Dealer	Via Dealer source chips; pairs with dealer business_name
color-brand-fallback (NEW)	#9E9E9E	Brand swatch fallback	Used for the brand colour swatch when brands.colour_hex is NULL (resolves OQ-10)
color-rating-filled (NEW)	#F57F17	Star rating, filled	Selected stars on SH-04 customer review form
color-rating-empty (NEW)	#D0D0D0	Star rating, empty	Unselected star outlines on SH-04

1.2 Typography

Typeface: Inter (or equivalent system sans-serif). System fallback stack: Inter, -apple-system, "Segoe UI", Roboto, sans-serif.

Token	Size	Weight	Usage
type-display	40 pt	Bold	Cover / marketing headers — not used in app UI
type-h1	18 pt	Bold	Page-level section titles (Dashboard, Analytics)
type-h2	15 pt	SemiBold	Component section headers within a page
type-h3 (CHANGED)	14 pt	SemiBold	Card titles, panel headings, modal titles. Raised from 13pt in v1.0 to create visual separation from type-body-lg
type-body-lg	13 pt	Regular	Primary body text, field labels, table rows
type-body-sm (CHANGED)	11 pt	Regular	Secondary labels, helper text, muted metadata. Raised from system floor of 10pt
type-label	11 pt desktop / 12 pt mobile	Medium / AllCaps	Chip / badge labels, status tags. Mobile minimum is 12pt to remain legible at xs in outdoor field conditions
type-code	11 pt	Monospace	UUID display, DB field references
type-caption (CHANGED)	11 pt	Regular	Timestamps, footnotes, tooltip text. Raised from 10pt — system-wide minimum is now 11pt to satisfy WCAG 1.4.4 floor

Mobile typography rule

On the xs breakpoint, type-label renders at 12pt minimum via responsive token override. The Status Chip component must respect this — see Section 2.

1.3 Spacing System

Base-4 scale. All layout spacing, padding, and margin values reference these tokens.

Token	Value	Primary Use
space-1	4 px	Inline gap between icon and label
space-2	8 px	Inner padding for chips/badges; compact list-item gap
space-3	12 px	Cell padding; button padding (vertical)
space-4	16 px	Card internal padding; form field margin

Token	Value	Primary Use
space-5	20 px	Panel internal padding (Technician PWA). Apply consistently to all interactive elements on the PWA, not only panels
space-6	24 px	Section spacing within a card
space-8	32 px	Gap between major layout panels
space-12	48 px	Page-level top/bottom margin; minimum touch target on the PWA (WCAG 2.5.5)
space-16	64 px	Drawer / modal top margin

⚠ iOS safe area

On the Technician PWA, the primary action button bottom padding must use CSS `env(safe-area-inset-bottom)` with a 16px fallback. Without this, CTAs are clipped on iPhones with a home indicator.

1.4 Elevation & Layout Rules

Layout Archetypes

Archetype	Used By	Structure
Full-width Dashboard	Owner, Office Staff	Top KPI card row → active job table → Amber Alert Panel. "Today's Focus" toggle (see §7) collapses the table to surface only KPI row + Amber Alert + Pending Schedule
Two-Panel Detail	Owner, Office Staff job detail	Left panel: job info + timeline. Right panel: action panel grouped into Workflow / Administrative / Payment sections separated by dividers
Scrollable Card List (PWA)	Technician	Tabs (Today / Upcoming / All) with badge counts → scrollable job card list → bottom action chip
Scoped Read-Only Portal	Dealer	KPI row → job history table → CTA header. Job detail uses simplified progress stepper. No sidebar nav
Step Form	Log New Job (OS-02, DL-02)	Stepper header → form body → Review & Submit footer. OS-02 auto-saves draft to localStorage on each step transition
Settings Form	Owner config, brand/dealer mgmt	Grouped sections → save button; single-column
Public Standalone	Customer review (SH-04)	Centred card, no navigation, branded header only

Elevation Tokens

Token	Box Shadow	Used For
elevation-0	none	Flat table rows, inline form fields
elevation-1	0 1px 3px rgba(0,0,0,0.12)	Cards, chips, notification items
elevation-2	0 4px 12px rgba(0,0,0,0.16)	Modals, overlay panels, bottom sheets, undo chip
elevation-3	0 8px 24px rgba(0,0,0,0.24)	Confirmation modals, blocking dialogs

Responsive Breakpoints

Breakpoint	Width	Target Context
xs	< 480 px	Technician PWA (mobile)
sm	480 – 768 px	Technician PWA (tablet) / Dealer portal mobile
md	768 – 1024 px	Dealer portal desktop; compact office views
lg	1024 – 1280 px	Primary office/owner desktop
xl	> 1280 px	Owner analytics, wide table views

1.5 Icon System (NEW)

Library: Lucide Icons. MIT-licensed, full React and SVG support, consistent 24px stroke-based geometry at all sizes. Default stroke width: 2. Default render size: 16px in chips and inline contexts; 20–24px in navigation, modal headers, and standalone affordances.

Status Group → Icon Assignments

Icons pair with the status colour and label as the third redundant signal required by WCAG 1.4.1. Within Status Chips, the icon renders at 14px, left-aligned, with a 4px gap (space-1) before the label.

Status Group	Lucide Icon	Status Values Covered
Scheduled states	Calendar	pending_schedule, scheduled
Assignment	UserCheck	assigned
Active progress	Wrench	acknowledged, in_transit, in_process
Completed / resolved	CheckCircle	completed, resolved, resolved_on_revisit
Needs attention	AlertTriangle	needs_revisit, revisit_scheduled

Status Group	Lucide Icon	Status Values Covered
Cancellation states	XCircle	cancellation_requested, cancelled
Complaint initial	MessageCircle	new
System events	Bot	All system_event timeline entries
Delayed sync	Clock	Delayed sync timeline events

Iconography elsewhere in the system

- Navigation: Home, ListTodo (jobs), Bell (notifications), Settings, BarChart3 (analytics), Users (technicians), Building2 (dealers).
- Affordances: Phone (call customer / office), MapPin (Navigate deeplink), Copy (copy ID), Share2 (share link), MessageCircle (WhatsApp action) paired with logo where contractually permitted.
- State icons: AlertTriangle (warning), AlertCircle (error), CheckCircle (success), Info (info banner), RefreshCw (sync in progress), WifiOff (offline).
- Modal headers: Lock (Action Required blocking modal), Undo2 (rollback), History (timeline), DollarSign (payment).

1.6 Animation Tokens (NEW)

Three tokens form the complete animation system. No additional tokens may be introduced without design sign-off.

Token	Value	Use
anim-instant	100 ms ease-out	Status chip re-render flash/pulse on WebSocket update; toast appearance
anim-standard	250 ms ease-in-out	Modal open/close, bottom sheet slide, overlay transitions, drawer expand
anim-countdown	1000 ms linear	Undo chip progress ring decrement — one tick per second

Undo chip implementation

Implement the countdown ring as a CSS stroke-dashoffset animation on an SVG circle, driven by anim-countdown and reset on each one-second tick — not as a single 60-second animation. This keeps the visual countdown accurate if the app is backgrounded and resumed mid-window. The visual timing is non-linear (slow for the first 40s, accelerating in the final 20s) while the wall-clock window remains exactly 60 seconds — see Component: 60-Second Undo Chip in §2.

2. Components

Component: KPI Card

Purpose: Displays a single aggregated business metric on the Owner and Office Staff dashboards. Cards are interactive and link to filtered job lists. They are the primary at-a-glance operational surface.

Variants

Variant	Data Source	Colour Treatment
Total Active Jobs	COUNT jobs WHERE status NOT IN terminal states	Neutral (primary blue)
Pending Schedule	COUNT WHERE status = pending_schedule	Accent blue; prominent if count > 0
Amber Alerts (Needs Revisit)	COUNT WHERE status = needs_revisit	Amber background when count > 0; green zero-state with checkmark when count = 0
Chronic Jobs	COUNT WHERE is_chronic = TRUE AND active	Red accent if count > 0
Today's Revenue	SUM analytics_business_daily WHERE date = TODAY	Currency formatted; green tint. Trend indicator MANDATORY
Completion Rate	Daily completion ratio	Percentage. Trend indicator MANDATORY
No-Show Count	analytics_business_daily.no_show_count	Neutral; links to filtered list

States

State	Visual Behaviour
Default	Shows metric value, label, and trend indicator (if applicable)
Zero (non-alert)	Metric shows 0. Card remains visible with neutral treatment
Zero (Amber Alert KPI)	Green zero-state: color-feedback-success-bg fill, CheckCircle icon (Lucide, green), metric value "0" in color-text-primary, label "No revisits pending". Amber treatment is reserved exclusively for count > 0
Loading	Skeleton shimmer replaces metric value; label stays visible. Stagger: Amber Alerts and Active Jobs load first; Today's Revenue loads second
Hover	Subtle elevation lift (elevation-1 → elevation-2); cursor: pointer

State	Visual Behaviour
Active / Clicked	Navigates to filtered job list view — no in-card toggle
Error	Shows '—' (em dash) with tooltip: 'Data unavailable'

Anatomy

- Metric Value (type-display or type-h1, prominent)
- Metric Label (type-body-sm, muted)
- Background fill (semantic per variant)
- Trend arrow + delta value — MANDATORY for Today's Revenue and Completion Rate; optional for other variants
- Click target: entire card surface

Behavior Rules

- DO: Clicking any KPI card navigates to a pre-filtered job list with filters matching the card's data condition. The applied filter is encoded in URL query parameters so back-navigation and bookmarks preserve it.
- DO NOT: Show disabled KPI cards for data the role cannot access — omit the card entirely.
- DO: Amber Alerts KPI uses amber background when count > 0 and the green zero-state when count = 0. The card is never hidden — its position is a navigational anchor.
- DO NOT: Use KPI cards outside dashboard contexts.
- DO: Render trend indicators (arrow + delta) for Today's Revenue and Completion Rate. The trend is the operationally important signal, not the absolute value.

Why a green zero-state, not a hidden card

Hiding the card on count = 0 creates layout shift each time alerts appear and disappear, which is disorienting on a dashboard scanned dozens of times daily. Staff who rely on spatial memory for the card's position lose their navigational anchor. The green zero-state with a checkmark is a positive operational signal that confirms all revisits are resolved.

Edge Cases

Analytics data unavailable

If analytics data is unavailable (e.g. DB timeout), show an em-dash placeholder and a 'Data unavailable' tooltip. Do not show 0 — this is misleading.

Acceptance Criteria

- ✓ KPI cards render within 2 seconds of dashboard load; skeleton shimmer is shown during load with the defined stagger.
- ✓ Clicking a KPI card navigates to a filtered job list with the correct filter pre-applied via URL query parameters.
- ✓ Amber Alert card displays amber background when count > 0 and the green zero-state when count = 0.
- ✓ Today's Revenue and Completion Rate render trend indicators on every load.
- ✓ Cards absent from roles that lack access — never shown as disabled.

Component: Status Chip

Purpose: Communicates the current job lifecycle state at a glance. Appears in all job-related surfaces: lists, cards, detail headers, timeline events, and the dealer portal (with label mapping). Status chips are the primary scanning surface for operational staff and must remain legible without colour interpretation.

Variants

Status Value	Display Label	Chip Colour Token	Lucide Icon	Notes
pending_schedule	Pending Schedule	color-status-pending (Grey)	Calendar	Dealer-submitted only
scheduled	Scheduled	color-status-scheduled (Blue)	Calendar	—
assigned	Assigned	color-status-assigned (Indigo)	UserCheck	Active assignment exists
acknowledged	Acknowledged	color-status-acknowledged (Teal)	Wrench	—
in_transit	In Transit	color-status-transit (Cyan)	Wrench	—
in_process	In Process	color-status-inprocess (Amber)	Wrench	—
completed	Completed	color-status-completed (Green)	CheckCircle	Terminal

Status Value	Display Label	Chip Colour Token	Lucide Icon	Notes
new	New	color-status-new (Grey)	MessageCircle	Complaint only
resolved	Resolved	color-status-completed (Green)	CheckCircle	Terminal
needs_revisit	Needs Revisit	color-status-revisit (#EF6C00) + 2px border	AlertTriangle	Amber alert also shown on dashboard
revisit_scheduled	Revisit Scheduled	color-status-scheduled (Blue)	AlertTriangle	Amber alert cleared
resolved_on_revisit	Resolved on Revisit	color-status-completed (Green)	CheckCircle	Terminal
cancellation_requested	Cancellation Pending	color-status-cancel-req (Red-light)	XCircle	All technician buttons disabled
cancelled	Cancelled	color-status-cancelled (Dark)	XCircle	Terminal; soft-delete NOT applied

States

State	Behaviour
Default	Filled chip with white label text; type-label (all caps recommended on desktop, sentence case fallback on xs)
Real-time update	Chip re-renders on WebSocket push or polling interval without full page reload AND emits a brief 100ms pulse animation (anim-instant token) to draw attention to the change
needs_revisit special	color-status-revisit (#EF6C00) chip + distinct 2px border; amber alert card raised on dashboard
Dealer-mapped label	Internal enum value translated to simplified label per Section 5 mapping before rendering in dealer portal

Anatomy

- Lucide icon (14px, left-aligned)
- 4px gap (space-1) between icon and label

- Background fill (semantic colour)
- Status label text (type-label, white, 11pt desktop / 12pt mobile)
- Optional: 2px border ring for needs_revisit
- Min-width: sufficient for longest label ('Cancellation Pending') without text wrapping

Behavior Rules

- DO: Render status chips everywhere a job is referenced — list rows, cards, detail headers, timeline events.
- DO: Always render the Lucide icon paired with the colour and the label. Icon + colour + label is the three-signal combination required for WCAG 1.4.1.
- DO NOT: Allow status chip to be an interactive element (no click behaviour on the chip itself).
- DO: Map internal enum values to simplified dealer labels in the dealer portal — never expose raw enum strings.
- DO NOT: Invent status values not defined in the spec.
- DO: Pulse on real-time status change (anim-instant). A chip that silently recolours mid-page will be missed or cause alarm.

Accessibility

- Icon + colour + text label are three redundant signals. Chip colour alone must NEVER be the only differentiator.
- Provide aria-label='Job status: [label]' on chip elements.
- type-label minimum size on xs is 12pt; sentence-case fallback is acceptable in place of all-caps when ALL CAPS becomes hard to scan at small sizes.

Acceptance Criteria

- ✓ All 14 job status values render with the correct icon + colour + label.
- ✓ needs_revisit chip uses #EF6C00 (color-status-revisit) and a distinct 2px border, visually separate from in_process which uses #F57F17.
- ✓ Dealer portal renders simplified label mapping — no raw enum values visible to dealer.
- ✓ Status chip updates without full page reload when WebSocket/poll delivers a status change, and emits a 100ms pulse on update.
- ✓ Chip text passes WCAG 1.4.1 — colour is never the sole differentiator.

Component: Job Card (Technician PWA)

Purpose: The primary unit of information in the Technician job list. Renders all job metadata needed to identify and prioritise a job before tapping into the action screen.

Variants

Variant	Context	Difference
Standard	Today / Upcoming / All tabs	Full metadata display
Revisit	Revisit jobs	Adds 'Revisit #N' badge; revisit sequence displayed
Frozen (Cancellation Pending)	cancellation_requested status	Grey overlay, no tap-through to action, with a defined message and contextual help — see States below
Completed / Terminal	completed, resolved, resolved_on_revisit	Read-only; action button absent

States

State	Visual Behaviour
Default	White card, elevation-1, all fields populated
Frozen	Grey overlay (rgba(0,0,0,0.25)); pointer events disabled for navigation; explicit message: "Cancellation pending — contact the office before proceeding." Cancellation request received timestamp shown beneath. One-tap Phone button (Lucide Phone icon) reaches the office directly. The phone affordance is the only interactive element on a frozen card
Offline	No change to card itself; persistent offline banner above list
Loading	Skeleton shimmer for card content
Empty list	'No assigned jobs — all caught up!' state with CheckCircle illustration

Anatomy

- Customer Name (type-h3 = 14pt SemiBold)
- Address (type-body-sm, muted), with a Navigate button (Lucide MapPin) that opens the device's native maps application
- Job Type chip (Installation / Complaint) — uses color-type-installation or color-type-complaint
- Status chip (icon + colour + label per chip convention)
- Brand name + colour_hex swatch circle (falls back to color-brand-fallback when null)

- Scheduled At — formatted 'Today 10:00 AM' or 'Mon 9 Jun'. Same-day jobs within 2 hours show a relative countdown in amber: "in 45 min". Jobs overdue by more than 30 minutes show "30 min ago" in red. Computed client-side from `scheduled_at`
- Revisit badge (conditional: 'Revisit #2')
- Review Pending indicator (conditional: when `review_mode` = mandatory and link generated but not submitted)

Behavior Rules

- DO: Tap on any non-frozen card navigates to the Job Action Screen (TC-02).
- DO NOT: Tap on frozen (`cancellation_requested`) card — overlay prevents navigation, except for the Phone affordance.
- DO: Show 'Review Pending' indicator only when `review_mode` = mandatory and link has been generated but not submitted.
- DO: Wrap the address in a deep link to the device maps application — single-tap navigation is the most common pre-acknowledge action and must not require copy-paste.
- DO: Support swipe-right to trigger the next logical status action (same as the primary action button) and swipe-left for a quick "Call Customer" action. For field technicians with gloves or adverse conditions, this reduces navigation depth.
- DO: Compute the urgency time-of-day signal client-side; do not depend on server polling for it.

Frozen card message — required copy

Headline: "Cancellation pending — contact the office before proceeding."

Sub-line: "Cancellation request received [relative timestamp]."

Action: Phone button labelled "Call office" (one tap, opens device dialer).

These are passive information actions that require no status change and can appear on a frozen card without unlocking it.

Acceptance Criteria

- ✓ Job cards with `cancellation_requested` status render with grey overlay, the defined frozen message, the cancellation-received timestamp, and a working call-office phone affordance.
- ✓ Tapping the body of a frozen card produces no navigation; tapping the call-office button opens the device dialer.
- ✓ Revisit badge shows correct sequence number pulled from DB.
- ✓ Brand colour swatch matches `brands.colour_hex` value exactly; falls back to `color-brand-fallback` when null.

- ✓ Scheduled time displays in local timezone, formatted per convention. Within-2-hour jobs show amber countdown; overdue >30 min jobs show red elapsed time.
- ✓ Address tap opens device maps; swipe-right triggers primary action; swipe-left triggers Call Customer.

Component: Primary Action Button (Technician)

Purpose: The single contextual action a technician can take on a job. Label and behaviour are entirely driven by the current job status. Appears at the bottom of the Job Action Screen.

Variants (by status)

Current Status	Button Label	Result
assigned	Acknowledge	status → acknowledged
acknowledged	In Transit	status → in_transit
in_transit	I've Arrived	status → in_process; actual_arrival = NOW()
in_process (installation)	Complete Job	Opens Payment Form Sheet (TC-03)
in_process (complaint)	Mark Resolved	Opens Payment Form Sheet (TC-03)
in_process (complaint)	Needs Revisit (secondary)	Opens Needs Revisit sheet (TC-04)
revisit_scheduled	Complete Job / Mark Resolved	Same as in_process flow
cancellation_requested	—	No button — frozen state message shown
completed / resolved / resolved_on_revisit	—	No button — read-only view

States

State	Visual
Default	Full-width, prominent fill button; type-h3 label; colour: color-primary-600. Bottom padding respects iOS env(safe-area-inset-bottom)
Loading / Submitting	Spinner replaces label; button disabled; prevents double-tap
Disabled (frozen)	Button absent entirely — replaced by explanatory message

State	Visual
Undo Active	Undo chip renders above the button after any transition; countdown visible (see global Undo Chip)

Behavior Rules

- DO: Render only one primary action button at a time — never multiple competing CTAs.
- DO: For complaint in_process, a secondary 'Needs Revisit' button may appear alongside 'Mark Resolved'.
- DO NOT: Show a disabled button for frozen/terminal states — omit entirely and show contextual message.
- DO: Button tap triggers optimistic local state update and queues the transition for sync.
- DO: Respect iOS safe-area-inset-bottom in CTA bottom padding to prevent home-indicator clipping. Use CSS `env(safe-area-inset-bottom)` with a 16px fallback.

Double-tap protection

The button must be disabled immediately on first tap and show a loading spinner until the transition is confirmed (locally queued or server-confirmed).

Acceptance Criteria

- ✓ Button label updates exactly with the status state machine — no mismatched labels.
- ✓ Tapping the button while offline queues the transition in the persistent device queue.
- ✓ 60-second undo chip appears above the button within 500ms of any successful transition.
- ✓ No action button is rendered for terminal or frozen states.
- ✓ Bottom padding incorporates `env(safe-area-inset-bottom)` and is not clipped on devices with a home indicator.

Component: 60-Second Undo Chip

Purpose: Provides a reversible window after any technician status transition. Prevents accidental, permanent state changes in a mobile PWA context where misclicks are common. The chip is a global sticky overlay anchored to the app shell — not a per-screen element — so the undo affordance is always reachable within the 60-second window regardless of navigation.

Variants

- Standard: displayed after any status transition by a technician.

- Post-completion: same chip after Complete Job / Mark Resolved; undo also rolls back the payment record (atomic).
- Post-needs-revisit: undo deletes the revisit row and clears the amber alert.

States

State	Visual
Active (counting down)	Floating compact pill (not full-width banner). Shape: pill, elevation-2. Label: 'Undo — [N]s'. Progress ring uses anim-countdown token, reset each second so the visual remains accurate after backgrounding. Visual progress is non-linear (slow for first 40s, accelerates in final 20s) — actual wall-clock window is 60s exactly
Tapped (undoing)	Chip shows 'Undoing...' spinner; chip disabled; rollback initiated
Expired	Chip auto-dismisses; transition is permanent
Offline during undo	Undo removes local queue entry if not yet synced; no server call needed
Synced before undo	Undo sends rollback request to server; payment deleted in same transaction if applicable

Anatomy

- Lucide Undo2 icon + 'Undo' label + countdown number + 's' suffix
- Progress ring rendered via SVG stroke-dashoffset, ticking each second using anim-countdown token
- Tap target: entire chip
- Position: global sticky overlay anchored to app shell; bottom of viewport; z-index above job cards but below modals; never blocks the primary action button area on TC-02

Behavior Rules

- DO: Show immediately (< 500 ms) after any technician status transition.
- DO: Persist across navigation within the 60-second window. The chip lives at the app-shell level so it remains visible on any PWA screen during the window.
- DO: On the job list (TC-01), tapping the chip initiates rollback without requiring navigation back to TC-02.
- DO NOT: Block swipe-to-action on cards directly behind the chip — the chip must remain compact and out of the swipe path.
- DO NOT: Allow chip to block the primary action button area on TC-02 — position above it with sufficient gap.

- DO NOT: Apply undo logic to Office Staff rollbacks — those use a separate modal flow with mandatory reason.

Edge Cases

⚠ Rollback semantics

Rolling back from needs_revisit via undo DELETES the revisit row (unlike staff rollback which leaves it as a historical record). Amber alert is cleared on undo but persists after the 60s window.

Undo on a Completed job rollback: payment row is deleted in the same transaction. The UI must confirm this behaviour is transactional to avoid payment orphan records.

Acceptance Criteria

- ✓ Countdown decrements accurately from 60 to 0 in real time, with non-linear visual progress and exact 60s wall-clock window.
- ✓ Chip is a global sticky overlay; visible on any PWA screen during the undo window; never obstructs swipe-to-action interactions.
- ✓ Undo on an unsynced transition removes the entry from device queue without a server call.
- ✓ Undo on a synced transition sends rollback and — for completed jobs — payment is deleted atomically.
- ✓ After 60 seconds, chip auto-dismisses; no further undo is possible.

Component: Amber Alert Panel

Purpose: Highlights jobs requiring follow-up (needs_revisit status) on the Owner and Office Staff dashboards. A dedicated surface that prevents high-priority jobs from being lost in the main job list.

Variants

- Dashboard Panel: full panel listing all needs_revisit jobs.
- Inline Card Tag: amber border/badge on job cards within any job list.
- KPI Card: Amber Alert count card in the KPI row.

States

State	Visual
Active (count > 0)	Panel shown; each row: Job ID, Customer, Brand, Technician, Last Visit Date, Revisit Sequence #
Chronic job in list	'CHRONIC' tag in color-feedback-error shown on panel row

State	Visual
Empty (count = 0)	Panel hidden in the body; the Amber Alert KPI card shows the green zero-state with checkmark and "No revisits pending" label. The KPI card itself is never hidden — its position is a navigational anchor
Job scheduled for revisit	Row removed from panel (status changes to revisit_scheduled; amber alert cleared)

Default Sort Order

1. CHRONIC jobs (is_chronic = TRUE) first.
2. Among CHRONIC jobs: secondary sort by revisit_sequence DESC.
3. Then non-CHRONIC needs_revisit jobs.
4. Among non-CHRONIC: sort by last_visit_date ASC (oldest first).

Sort control in the panel header allows staff to switch to alternate views (newest first, by technician, by brand). Default order surfaces the most urgent items at the top of the panel.

Behavior Rules

- DO: Show CHRONIC tag on rows where is_chronic = TRUE.
- DO: Default sort surfaces CHRONIC jobs above non-CHRONIC, oldest unresolved on top.
- DO NOT: Show this panel in the Technician or Dealer portals.
- DO: Panel rows should be clickable — navigate to the job detail.
- DO: When a cancellation_requested job has been pending more than 30 minutes without staff action, render the cancellation request as a persistent amber banner at the top of the dashboard (above the Amber Alert Panel) with inline Approve/Reject CTAs. This prevents cancellation requests being buried during busy periods.

Acceptance Criteria

- ✓ Panel renders all jobs WHERE status = needs_revisit, default-sorted CHRONIC-first then oldest first.
- ✓ CHRONIC tag is shown on rows where is_chronic = TRUE.
- ✓ Panel body is hidden when count = 0; the corresponding KPI card shows the green zero-state.
- ✓ Stale cancellation requests (>30 min) escalate to a persistent dashboard banner with Approve/Reject CTAs.
- ✓ Navigating from a panel row opens the correct job detail.

Component: Modal / Confirmation Dialog

Purpose: Used for high-stakes, irreversible, or role-exclusive actions that require deliberate user confirmation with a mandatory reason. Never used for simple notifications.

Variants

Variant	Trigger	Contents
Status Override (Owner only)	Owner clicks 'Full Status Override'	Status dropdown (all valid for job type) + mandatory reason textarea (≥10 chars) + Confirm button. Collapsible "What does this mean?" help section
Reopen Cancelled Job (Owner only)	Owner clicks 'Reopen Job' on cancelled job	Mandatory reopen reason textarea (≥10 chars) + Confirm button
Void-or-Retain Payment (Owner only) — BLOCKING	Owner reverses a completed/resolved job outside 60s window	Two mutually exclusive options: Void (+ void_reason ≥10 chars) or Retain (+ retained_note). Each option carries a one-line description (Void: "Remove this payment record entirely" / Retain: "Keep the payment record without reversing it"). Cannot be dismissed without a choice
Rollback Status (Staff)	'Roll Back Status' on pre-completion jobs	Mandatory reason textarea (≥10 chars). Tooltip on the trigger button shows target state before opening: "Roll back to: In Transit"
Direct Cancellation (Owner / Staff)	'Cancel Job' from job detail	Optional in-process warning → mandatory cancellation reason textarea (≥10 chars) + Confirm
Cancellation Request Review (Staff / Owner) — BLOCKING	Job has status = cancellation_requested	Dealer reason (read-only) + Approve / Reject buttons — no dismiss without action
Reassignment Warning	Reassigning technician who is in_transit	Warning text + Continue / Cancel buttons
Scheduling Conflict	Technician has an overlapping job in window	Warning + Acknowledge & Proceed / Select Different Technician. If staff regularly tolerate routine double-bookings, evaluate downgrading this to a dismissible warning toast

Distinguishing Blocking from Dismissible Modals

Users conditioned to dismiss modals via ESC or backdrop click will experience blocking modals as a silent system failure unless the constraint is communicated visually.

Treatment	Dismissible Modal	Blocking Modal (Void-or-Retain, Cancellation Request Review)
Close affordance	✕ button visible in header	✕ button REMOVED entirely (not disabled — absent)
Header badge	None	"Action Required" badge (Lucide Lock icon) in modal header
ESC keypress	Closes modal	Triggers brief shake animation (300ms, ±4px horizontal) + inline tooltip: "You must make a selection to continue."
Backdrop click	Closes modal	Same shake-and-tooltip response as ESC
Submit button	Enabled when valid	Submit/choice buttons are the only path forward

States

State	Behaviour
Default	Modal centred, backdrop dimmed (elevation-3), focus trapped inside, anim-standard open transition
Reason required (empty)	Submit button disabled; reason field shows required asterisk; counter shows 'X / 10 minimum'
Reason filled (≥10 chars)	Submit button enabled
Submitting	Submit button shows spinner; all inputs disabled; backdrop remains
Success	Modal closes (anim-standard); parent screen updates (toast or inline confirmation)
Error	Field-level error shown inside modal; modal stays open for correction
Cannot dismiss (blocking)	Per the Distinguishing table above: ✕ absent, Action Required badge present, ESC/backdrop click → shake + tooltip

Behavior Rules

- DO: Trap focus within modal when open (accessibility).
- DO: Disable submit button until all mandatory fields pass validation, including the 10-character reason minimum.
- DO: Void-or-Retain modal MUST intercept status reversal before the status update is applied. The status update must not proceed if the modal is bypassed.

- DO: For low-frequency, high-stakes modals (Void-or-Retain, Status Override, Reopen Cancelled), include a collapsible "What does this mean?" help section. First-time users will be uncertain — particularly for Void vs Retain which has financial consequences.
- DO NOT: Allow blocking modals (Void-or-Retain, Cancellation Request Review) to be dismissed via backdrop click or ESC key. ESC and backdrop click must produce a shake animation and tooltip rather than silent inaction.
- DO NOT: Pre-select a default option in Void-or-Retain — force explicit user choice.
- DO NOT: Open the Roll Back Status modal without first showing a tooltip on the trigger that names the target state.

Edge Cases

⚠ Void-or-Retain dismissal

This modal cannot be dismissed without making a choice. The underlying status reversal is blocked until Void or Retain is selected. ESC and backdrop click respond with shake animation + tooltip — not silent rejection.

⚠ Reopen Cancelled Job — pre-cancel state lookup

The backend must query `job_timeline` to find the last non-cancelled status. If none exists, show error inside modal: 'Cannot determine pre-cancellation status — contact support.' Confirm button must be disabled in this state.

Acceptance Criteria

- ✓ Submit button remains disabled until all required modal fields pass validation, including 10-character reason minimum.
- ✓ Void-or-Retain modal cannot be closed by ESC, backdrop click, or any method other than selecting Void or Retain — and ESC/backdrop click trigger the shake animation and tooltip.
- ✓ Cancellation Request Review modal forces Approve or Reject — no neutral dismiss available.
- ✓ Status override and rollback actions write the correct timeline event type to the database.
- ✓ Roll Back Status trigger button shows target-state tooltip before opening the modal.
- ✓ Void-or-Retain options each carry a one-line description; the collapsible help section is present.

Component: Form Field System

Purpose: Standardises all input, validation, and error display across the Log New Job form, configuration screens, dealer management, and any other data-entry surface.

Variants

Type	Used For
Text input	Customer name, address, brand name, business name
Email input	Dealer account email, login
Tel input	Customer phone (triggers async VCID lookup on blur)
Password input	Login, initial dealer password (show/hide toggle; strength indicator on creation)
DateTime picker	scheduled_at — must enforce future datetime; local timezone display
Select / Dropdown	Brand, Technician, Payment Method, Job Type, Status (override)
Multi-select	Brand Assignment (dealer), filter chips in job list
Radio group (desktop only)	Job Type (installation/complaint), Job Source (direct/via_dealer). On the Technician PWA, replace with full-width tappable chips/cards (≥48px height) — see TC-04 in §4
Textarea	Issue description, cancellation/void/rollback/withdrawal reasons. Reason textareas enforce a 10-character minimum (resolves OQ-07)
Currency input	Payment amount (decimal keyboard on mobile; amount ≥ 0 validation)
Toggle	is_active for dealers, payment methods
Repeating row	Unit details (model, unit type, num units); add/remove row buttons; minimum 1 row
Star selector (SH-04)	Customer review submission. 5 stars; 48px star size; 8px gap; selected = filled #F57F17 (color-rating-filled); unselected = grey outline #D0D0D0 (color-rating-empty); press state = scale(1.1) transform. Required before text

States (all field types)

State	Visual / Behaviour
Default	Label + input; required fields marked with red asterisk (*)
Focused	Border highlight (color-primary-600); no label change
Filled (valid)	Input shows value; no error indicator
Error (client-side)	Red border; error message below input (fires on blur, not keystroke)
Error (server-side)	Same visual as client error; message reflects DB constraint or service rejection
Disabled	Greyed out; not focusable; cursor: not-allowed

State	Visual / Behaviour
Loading (async lookup)	Spinner inside field or adjacent (phone VCID lookup)
Read-only	Text displayed without input chrome; no edit affordance (dealer view)

Validation Rules (Global)

- Inline validation fires on blur — never on every keystroke.
- Submit button is disabled until all required fields pass client-side validation.
- Cancellation, void, rollback, and withdrawal reason fields are mandatory AND require a 10-character minimum. The character counter shows 'X / 10 minimum' beneath the textarea while below threshold.
- Server-side errors surface as field-level messages below the relevant input.
- Brand selector: if selected brand is inactive, submission is blocked with error: 'This brand is inactive. Select an active brand to continue.'
- Dealer selector (via_dealer source): if no dealer_id is set, submit remains disabled.

Phone Lookup Behaviour (VCID)

The phone field in OS-02 (Log New Job) triggers an async VCID resolution on blur. Confidence is communicated explicitly — staff should not need to infer it from layout.

Outcome	UI Behaviour
Match Found (primary)	Solid suggestion card with header label "Best match" — shows '[Name] — [Address]' with Confirm / Dismiss buttons
Multiple VCIDs matched	Both matches shown as separate "Best match" cards; staff must select one or explicitly create a new record. Cannot proceed without a choice
Partial match (Levenshtein)	Dashed-border suggestion card with header label "Possible match" + numeric confidence (e.g. "87% match"); same Confirm/Dismiss controls
No match	No suggestion shown; proceeds to new VCID generation



Dealer-side phone lookup

Per OQ-04 (resolved): dealers see only "New Customer" / "Existing Customer" — no VCID details, no name disclosure, no confidence indicators. The dealer's responsibility is to submit accurate contact information; conflict resolution is internal staff work.

Acceptance Criteria

- ✓ Inline errors fire on blur — not on keystroke.
- ✓ Submit button is disabled until all required fields pass validation, including the 10-character reason minimum.
- ✓ Phone lookup shows spinner during async resolution; suggestion card renders with explicit "Best match" / "Possible match" labelling and percentage for partial matches.
- ✓ Repeating row (unit details): add/remove row buttons work correctly; at least 1 unit row is required for submission.
- ✓ Star selector renders at 48px star size with the defined fill and press states.

Component: Notification Centre

Purpose: Central panel for in-app notifications. Replaces push notifications entirely. Badge count drives urgency; read/unread state drives discoverability.

Variants

- Nav badge: unread count on bell icon in navigation bar.
- Panel: slide-in or dropdown panel showing all notifications newest-first, with grouping and pagination.
- Notification item: individual notification card within the panel.

Notification Types

Type	Recipients	Urgent Styling
dealer_job_submitted	Owner, Office Staff	No
cancellation_request_outcome	Dealer	No
status_changed	Owner, Office Staff (configurable)	No
job_assigned	Technician	No
needs_revisit_raised	Owner, Office Staff	Yes (amber accent)
pre_chronic_warning (NEW)	Owner	Yes (amber accent). Fires at revisit_sequence = 2: "This job has had 2 revisits. One more will flag it as Chronic." Gives owner an actionable window to intervene before the threshold is crossed
chronic_flagged	Owner	Yes (red accent)

Type	Recipients	Urgent Styling
low_rating_received (NEW)	Owner, Office Staff	Yes (red accent). Fires when <code>is_low_rated = TRUE</code> ; payload includes star rating, job reference, technician name. Surfaced as a distinct service-recovery item
cancellation_auto_rejected (NEW)	Owner, Office Staff, Dealer	No (informational system event)

States

State	Visual
Unread	Blue dot on item; item background slightly elevated
Read	Blue dot absent; item dimmed
Panel loading	Skeleton shimmer on item list
Empty	'You're all caught up' with check icon (CheckCircle, Lucide)
Badge: unread count > 0	Numeric badge on nav bell icon
Badge: 0 unread	Badge hidden (no '0' badge)

Grouping, Filtering, Pagination

- Group notifications by `job_id` when two or more relate to the same job. The group renders as a single collapsible item: "3 updates on Job #AB123412" — expandable on tap to show the constituent items.
- Panel header includes a filter chip row: All | Unread | Cancellations | Assignments. Selecting a chip narrows the rendered list; URL query param persists the choice across sessions.
- Render with virtual scrolling or paginate at 50 items. In high-volume operations a flat unbounded list is unusable.

Behavior Rules

- DO: Clicking a notification (or an expanded group item) marks `is_read = TRUE` and `read_at = NOW()` and navigates to the related job.
- DO: 'Mark all as read' action available at panel header.
- DO: Render urgent-styling variants (`low_rating_received`, `pre_chronic_warning`, `chronic_flagged`, `needs_revisit_raised`) with their distinct accent colour.

- DO NOT: Implement client-side deduplication — backend handles via 4 partial indexes.
- DO NOT: Show notifications intended for other roles (each role sees only their role-targeted events).
- Dealer portal: shows only cancellation_request_outcome events addressed to that dealer.

Acceptance Criteria

- ✓ Unread badge count increments for each new unread notification; decrements when read.
- ✓ Multiple notifications on the same job render as a single grouped item, expandable on tap.
- ✓ Filter chip row supports All / Unread / Cancellations / Assignments and persists in URL.
- ✓ Panel paginates or virtualises at 50 items.
- ✓ low_rating_received and pre_chronic_warning render with their distinct urgent styling.
- ✓ Clicking a notification marks it read and navigates to the correct job.
- ✓ Technicians do not see owner/staff targeted notifications and vice versa.
- ✓ Empty state renders 'You're all caught up' when no notifications exist.

Component: Job Timeline Viewer

Purpose: Provides a full audit trail of all status changes, payment edits, scheduling events, and system-generated events on a job. Used by Owner and Office Staff only.

Variants

- Standard event: status change, assignment, scheduling, cancellation, rollback.
- System event: backend-generated events (auto-rejection, conflict logging); distinct Bot icon (Lucide) and muted colour.
- Delayed sync event: shows Clock icon (Lucide); occurred_at reflects client-reported timestamp.

System Event Types

event_type	Actor	When written
scheduling_conflict_logged	System	Staff acknowledged a scheduling conflict during assignment
cancellation_auto_rejected	System	Stale cancellation request rejected because the job lifecycle advanced. Reason: "Job lifecycle advanced beyond cancellation request." Written atomically within the same transaction as the causing status transition (see §3 Pattern: Cancellation Flow)

event_type	Actor	When written
chronic_threshold_reached	System	is_chronic flag flipped to TRUE on revisit_sequence = 3
review_link_expired	System	Customer review token expired without submission

Anatomy (per event card)

- event_type: human-readable label + category icon (Lucide)
- occurred_at: relative time (e.g. '2 hours ago') with absolute datetime on hover
- Actor: full_name (user), business_name (dealer), or 'System' for system_event
- before/after diff: shown when previous_value and new_value both present; scheduled_at changes formatted as human-readable datetime diff
- reason: shown for rollback, cancellation, void events

Behavior Rules

- DO: Render newest event at top.
- DO: System events use muted text colour and the Bot icon to distinguish from human actions.
- DO NOT: Allow editing of any timeline entry — read-only.

Acceptance Criteria

- ✓ Timeline renders all job_timeline rows for a job, ordered newest-first.
- ✓ System events render with the Bot icon and muted colour; cancellation_auto_rejected events show actor = System and the canonical reason string.
- ✓ before/after diff is shown when both previous_value and new_value are present.
- ✓ reason field is shown for rollback, cancellation, and void events.

3. Patterns

Pattern: Job Creation Flow

Trigger: User clicks 'Log New Job' (Office Staff / Owner) or 'Submit New Job' (Dealer portal).

Flow Logic (Office Staff / Owner — OS-02)

5. Step 1 — Job Type & Source: Select Installation or Complaint. Select Direct or Via Dealer. If Via Dealer, Dealer selector appears; `chk_dealer_source` enforced.
6. Step 2 — Customer Identity Resolution: Enter primary phone. System performs async VCID lookup on blur. Handle 4 match states with explicit confidence labelling (see Form Field component). Staff confirms or dismisses suggestion.
7. Step 3 — Job Details: Fields rendered based on job type (installation: Brand, Scheduled At, Unit Details; complaint: Brand, Issue Description, Unit Details). For repeat customers (VCID matched in Step 2), the customer's most-recently-used brand is offered as a one-tap suggestion chip: "Last brand: [X] — use again?" The chip is a suggestion only — never a default.
8. Step 4 — Review & Submit: Summary panel of all values. 'Create Job' button. On success: navigate to new job detail. On failure: surface field-level server errors.

Draft Auto-Save Across Steps

Form state is auto-saved to `localStorage` on each step transition. On returning to OS-02, a "Resume Draft" indicator at the top of Step 1 shows the customer name and timestamp: "Resume: [Name] — started 14 min ago." The draft is cleared on successful submission or explicit "Discard" action. This prevents data loss in the call-centre environment where mid-form interruptions are constant.

Flow Logic (Dealer — DL-02)

9. Single-page form: Job Type, Brand (scoped to `dealer_brands`), Customer Name, Customer Phone, Address, Model, Unit Type, Unit Details/Issue Description.
10. Phone lookup runs internally; dealer sees only 'New Customer' or 'Existing Customer' — no VCID details, names, or confidence indicators exposed (resolves OQ-04).
11. On success: in-page success state shows reference ID + two CTAs — primary "View Job" navigating to DL-03 for the new job, secondary "Submit Another". Owner + all office staff receive `dealer_job_submitted` notification.

UI Behavior

- Step form or single-page form — job type cannot be changed after Step 1 without resetting the form.

- Installation with no scheduled_at: if Direct source, this is an error (must schedule at creation). If Dealer source, status = pending_schedule.
- Complaint submission: status = new.

Failure Handling

Brand inactive: block submission; show field error on brand selector.

Via Dealer with no dealer_id: submit disabled; field-level error on dealer selector.

Job type mismatch (server rejects): "Something went wrong on our end. Your form data is saved — please try again." (Replaces the v1.0 "Job type mismatch. Please reload and try again." copy.)

Phone linked to two VCIDs: both shown as "Best match" cards; staff must select one manually.

Network failure (Dealer): no partial INSERT committed. Show: "Your job couldn't be submitted due to a connection issue. Your form data has been saved. Tap Retry to try again." Show a retry counter if multiple attempts fail.

Acceptance Criteria

- ✓ Job type cannot be changed after Step 1 without form reset.
- ✓ Direct installation with no scheduled_at fails validation before submission.
- ✓ Dealer job with no dealer_brands match cannot proceed (brand absent from dropdown).
- ✓ OS-02 auto-saves the in-progress form to localStorage on each step transition; "Resume Draft" surfaces on next visit with customer name and elapsed time.
- ✓ Repeat-customer brand suggestion chip appears in Step 3 when VCID matched and a prior brand exists; selecting it pre-fills the brand without auto-applying it.
- ✓ DL-02 success state offers "View Job" and "Submit Another" CTAs — never a dead end.
- ✓ On success, user is navigated to the new job detail screen with the correct job loaded.

Pattern: Status Transition System

Trigger: Technician taps primary action button (TC-02); Office Staff or Owner clicks 'Advance Status', 'Roll Back Status', or 'Full Status Override' in the action panel.

State Machine

The following transitions are valid per role:

From	To	Who	Method
assigned	acknowledged	Technician	Primary action button tap
acknowledged	in_transit	Technician	Primary action button tap

From	To	Who	Method
in_transit	in_process	Technician	Primary action button tap; records actual_arrival
in_process (installation)	completed	Technician	Complete Job → Payment Confirmation → Submit
in_process (complaint)	resolved	Technician	Mark Resolved → Payment Confirmation → Submit
in_process (complaint)	needs_revisit	Technician	Needs Revisit sheet → Submit
needs_revisit	revisit_scheduled	Owner / Staff	Pending Schedule Queue or Job Detail
revisit_scheduled	in_process (revisit)	Technician	Primary action button tap
any non-terminal	cancelled	Owner / Staff	Cancel Job modal with mandatory reason
any active	cancellation_requested	Dealer	Request Cancellation form
cancellation_requested	cancelled	Owner / Staff	Approve in Cancellation Request Review
cancellation_requested	prior status	Owner / Staff	Reject in Cancellation Request Review
cancelled	last pre-cancel status	Owner only	Reopen Job modal
any	any (valid)	Owner only	Full Status Override modal
One state back (limited set)	previous state	Owner / Staff	Roll Back Status modal (tooltip on trigger names target state before opening)

UI Behavior

- Technician: transition is optimistic — state updates locally, queued for sync. Global sticky undo chip shown for 60 seconds across any PWA screen.
- Staff/Owner rollback: one-state only (pre-completion). Mandatory reason (≥10 chars). No undo chip — not reversible by technician. Tooltip on the rollback button names the target state before the modal opens.
- Owner override: any valid status; mandatory reason; writes status_rollback timeline event.
- Status chip updates in real-time via WebSocket or polling without page reload, with a 100ms pulse (anim-instant) on update.
- Staff rollback from needs_revisit: revisit row is preserved as a historical record (resolves OQ-03). The 60s technician undo deletes it; staff rollback does not.

Failure Handling

⚠ HTTP 409 (version conflict)

Show non-blocking toast: "This record was updated by someone else. Refreshing..." Auto-fetch latest version.

If user had pending changes in-flight: open a bottom sheet (not modal) showing "Your changes" vs "Current version" in a two-column diff view, with three options: Keep mine / Use latest / Review both. Do not silently discard the user's changes.

⚠ Server rejects a queued technician transition

If a queued transition is rejected after sync (e.g. job cancelled while offline), show error toast with the specific rejection reason. The queue item stays for manual resolution; the offline sync queue indicator changes to a red error state. Do not silently decrement the count.

Acceptance Criteria

- ✓ All valid transitions listed in the state machine succeed when performed by the correct role.
- ✓ Transitions by disallowed roles are not rendered — no disabled button; controls absent.
- ✓ Status chips emit a 100ms pulse on real-time update.
- ✓ Roll Back Status trigger button shows a tooltip naming the target state before the modal opens.
- ✓ 409 conflict shows toast and, when there are pending changes, opens a 3-way diff bottom sheet.
- ✓ Offline transition queues atomically and syncs on connectivity restore; rejected queue items surface a red error state rather than silently disappearing.

Pattern: Technician Assignment & Scheduling Conflict Check

Trigger: 'Assign Technician' or 'Reassign Technician' button on Job Detail; inline assignment in Pending Schedule Queue (OS-04); Batch Schedule mode in OS-04.

Flow Logic — Single Assignment

12. Open technician picker (modal or right-panel): dropdown of users WHERE role = technician AND is_active = TRUE AND is_deleted = FALSE. Search/filter supported.
13. If this is a complaint and the customer has a prior completed job, the picker pre-suggests the original technician at the top of the list with the label "Assign to [Name] — completed prior installation for this customer." Familiarity reduces complaint resolution time. The suggestion is never auto-applied.

14. On selection: system queries for scheduling overlap within standard_job_duration_mins window (from system_config).
15. No conflict: Confirm button enabled. Assignment proceeds on confirm.
16. Conflict detected: warning banner shown with conflicting job ID. Two options: (a) Acknowledge & Proceed — logs scheduling_conflicts row with canonical pair ordering. (b) Select Different Technician — returns to picker, no conflict logged.
17. Reassignment: previous assignment marked is_active = FALSE; new row inserted. Previous technician loses job from view immediately.

Flow Logic — Batch Scheduling (OS-04)

In peak periods, staff may have many dealer-submitted jobs awaiting scheduling. Batch mode turns a 15-action workflow into a 3-action one.

18. Toggle "Batch Schedule" in OS-04. Checkbox column appears on each pending row.
19. Select multiple pending jobs → "Assign to Technician" → choose technician from the picker.
20. System auto-spaces selected jobs based on standard_job_duration_mins, shows a preview of the proposed schedule (start times, gaps, conflicts highlighted), and allows time adjustment per row before final confirm.
21. Single "Confirm Schedule" action commits all assignments atomically.

Pending Schedule Queue — Days Waiting Threshold

Resolves OQ-02. The Days Waiting counter on each row is highlighted using two configurable thresholds:

Job Type	Soft Threshold (Amber)	Hard Threshold (Red)
Installation	2 days	4 days
Complaint	1 day	2 days

Both thresholds are exposed in OW-04 (System Config) as pending_schedule_amber_days and pending_schedule_red_days, with separate values per job type.

Failure Handling

⚠ Reassignment while technician is in_transit

System permits it but UI must show confirmation: 'This technician is currently in transit to this job. Reassigning will remove it from their view. Continue?'

Acceptance Criteria

- ✓ Only active, non-deleted technicians appear in the picker.
- ✓ Repeat-complaint suggestion appears at the top of the picker when a prior completed job exists for this customer.
- ✓ Conflict warning is shown before confirmation — not after.
- ✓ Canonical pair ordering (LEAST/GREATEST) is enforced — no duplicate conflict rows.
- ✓ Previous technician's job list updates immediately on reassignment.
- ✓ Batch Schedule mode in OS-04 supports multi-select, auto-spacing preview, per-row adjustment, and atomic commit.
- ✓ Days Waiting cell is highlighted amber/red per the configured thresholds.

Pattern: Cancellation Flow

Trigger: Owner/Staff click 'Cancel Job'; Dealer clicks 'Withdraw' or 'Request Cancellation'.

Flow Logic — Direct Cancellation (Owner / Staff)

22. Allowed when job is in any non-terminal, non-completed state.
23. If status = in_process: show intermediate confirmation 'Work has commenced on this job. Confirm cancellation?' before showing reason field.
24. Mandatory cancellation reason textarea (≥10 chars). Submit disabled until threshold met.
25. On confirm: status → cancelled. Job excluded from operational queues but visible in history.

Flow Logic — Dealer Withdrawal (Initial State Only)

26. Only when status = new (complaint) or pending_schedule (installation).
27. Mandatory withdrawal reason (≥10 chars — resolves OQ-07). 'Withdraw Job' confirms.
28. Status → cancelled. Shows as 'Withdrawn' in dealer portal.

Flow Logic — Dealer Cancellation Request (Beyond Initial State)

29. Dealer submits mandatory reason (≥10 chars) → status → cancellation_requested.
30. All technician action buttons frozen. The technician's Job Card renders with the defined frozen message and call-office affordance — see Job Card component.
31. DL-04 in-page success state: "Your cancellation request has been submitted. You'll be notified once a decision is made." Includes the job reference.
32. Staff/Owner see Cancellation Request badge on job action panel as a persistent attention banner (amber background, dealer reason in collapsed form, Approve/Reject CTAs below) — not a competing button among others.

33. If the request is not actioned within 30 minutes, it escalates to a persistent dashboard banner with inline Approve/Reject CTAs (see Amber Alert Panel behaviour).
34. Approve → cancelled. Reject → job returns to held status; technician buttons re-enabled.

Auto-Rejection (resolves OQ-05)

If the job lifecycle advances past a state where cancellation is meaningful while a `cancellation_requested` row is still pending, the request is auto-rejected. The trigger is a service-layer check on every status transition — NOT a background job.

- The auto-rejection writes a `system_event` row of type `cancellation_auto_rejected`, actor = System, reason = "Job lifecycle advanced beyond cancellation request."
- Both events (the status transition and the auto-rejection) are written in the same DB transaction. Timeline timestamp is exact and matches the causal transition; WebSocket push fires immediately for dealer and staff; the technician's buttons are re-enabled in the same operation; no window exists where the system is in an inconsistent state visible to any role.
- Dealer portal: notification "Cancellation Rejected — job has progressed." Job returns to active status.

Failure Handling

Dealer attempts second cancellation request while one is still pending: submit disabled. Message: 'A cancellation request is already pending for this job.'

Staff cancelling a completed/resolved job: Owner access required. Staff button absent; message reads 'Status reversal for closed jobs requires Owner access.'

Acceptance Criteria

- ✓ Cancellation reason field is mandatory and enforces a 10-character minimum.
- ✓ Withdrawal reason field also enforces the 10-character minimum (resolves OQ-07).
- ✓ Direct cancellation of `in_process` job shows intermediate confirmation before reason entry.
- ✓ Dealer cannot submit a second cancellation request while one is pending.
- ✓ Cancellation Request Review forces Approve or Reject — no modal dismiss without action.
- ✓ DL-04 shows in-page submission confirmation with reference ID.
- ✓ Auto-rejection fires from the service-layer check; writes `cancellation_auto_rejected` system event with reason "Job lifecycle advanced beyond cancellation request".
- ✓ Cancellation Request panels render as persistent attention banners; stale requests (>30 min) escalate to a dashboard-level banner.

Pattern: Payment Collection & Management

Trigger: Technician taps 'Complete Job' or 'Mark Resolved'; payment management actions on Job Detail by Owner/Staff.

Flow Logic — Collection (Technician)

35. Payment Form Sheet opens (TC-03): amount (≥ 0 , decimal) + payment_method_id (active methods only).
36. Technician taps Submit → one-screen confirmation summary appears: "Complete [Job Type] — Amount: [X] — Method: [Y]. Confirm?" with large Confirm and Back buttons. This single additional tap prevents the most costly technician error path; an incorrect amount or wrong payment method otherwise creates an owner-level void/retain workflow to correct.
37. Confirm commits status transition + payment as a single atomic queue entry. Both written in one DB transaction (status → completed/resolved; payment row INSERTed as 'collected').
38. Payment is inserted as 'collected' directly — no pending intermediate state.
trg_set_collected_at fires on INSERT.
39. After submission: technician cannot modify payment. No edit controls shown.

Flow Logic — Management (Owner / Staff)

Action	Who	Conditions
Edit Payment Method	Owner, Staff	Job active (not terminal). Writes payment_edited timeline event
Edit Payment Amount	Owner only	Payment exists and job not in terminal state via standard path. Writes payment_edited event
Void Payment	Owner only	Owner reverses completed/resolved status outside 60s window AND payment exists. Mandatory void_reason (≥ 10 chars). Triggered via Void-or-Retain blocking modal
Retain Payment	Owner only	Same trigger as Void. Updates retained_note. No amount change. Triggered via Void-or-Retain blocking modal

Failure Handling

Offline payment submission: payment + status queued together as atomic entry; synced on connectivity restore.

Duplicate payment on retry: UNIQUE constraint on payments.job_id. Server returns 409 — client treats as success (payment already recorded).

Void-or-Retain modal: cannot be dismissed without a choice. Status reversal is blocked until choice is made. ESC and backdrop click trigger the shake animation and tooltip — see Modal component.

Acceptance Criteria

- ✓ Technician sees a one-screen confirmation summary before final submission.
- ✓ Payment submission and status transition are atomic — one fails, both fail.
- ✓ Duplicate payment retry (409) is treated as success, not an error, in the technician UI.
- ✓ Technician sees no edit controls on payment after submission.
- ✓ Void-or-Retain modal intercepts status reversal before it is applied; ESC and backdrop click trigger shake-and-tooltip rather than silent failure.

Pattern: Customer Review Flow

Trigger: Status transition to completed or resolved_on_revisit AND customer_review_mode ≠ off.

Flow Logic

40. After completion transition, review sheet opens (TC-05).
41. Optional mode: Two buttons — primary 'Share via WhatsApp', secondary 'Copy Link', and 'Skip'. Tap-outside or device back also count as Skip in optional mode (resolves OQ-08). Skipping creates no customer_reviews row; job fully closed.
42. Mandatory mode: 'Skip' absent; tap-outside and back-navigation are blocked. Technician must generate the link. 'Review Pending' indicator on job card until customer submits or 48h expires.
43. On generate: review_token (UUID) created; expires_at = link_generated_at + 48h. Shareable URL shown with primary 'Share via WhatsApp' CTA, secondary 'Copy Link', and 'Share via SMS' deeplink.
44. Customer opens link (SH-04): selects star rating (1–5, required, 48px touch targets), optional text. Submits. Page validates expires_at before rendering form.
45. On submit: customer_reviews updated. is_low_rated generated column auto-sets ($\leq 2 = \text{TRUE}$). low_rating_received notification fires for Owner and Office Staff with urgent styling (see Notification Centre).



Why WhatsApp is the primary CTA

In the product's primary markets WhatsApp is the dominant messaging channel. Listing Copy Link and WhatsApp as co-equal CTAs adds an unnecessary decision step. Visual hierarchy: WhatsApp

primary (filled button, color-primary-600), Copy Link secondary (outline button), SMS tertiary (text link).

Failure Handling

Expired token: 'This review link has expired.' — no form rendered.

Invalid token: 'This review link is not valid.' — no further information shown.

customer_review_mode set to 'off' while reviews in progress: indicator clears on already-completed jobs. No retroactive change.

Acceptance Criteria

- ✓ Mandatory mode: 'Skip' button is absent; tap-outside and back-navigation are blocked; technician cannot close sheet without generating link.
- ✓ Optional mode: tap-outside and back-navigation count as Skip and close the sheet without generating a link (resolves OQ-08).
- ✓ Share via WhatsApp is the primary CTA; Copy Link is secondary.
- ✓ Review link expires at link_generated_at + 48h; expired links show correct message, no form.
- ✓ is_low_rated = TRUE auto-set for ratings ≤ 2 via generated column. low_rating_received notification fires with urgent styling.
- ✓ avg_star_rating column in analytics is hidden when customer_review_mode = off.

4. State & Interaction System

4.1 Global State Rules

State	Visual Pattern	Behaviour
Loading	Skeleton shimmer on data regions; spinner on buttons during submission	Non-blocking for data; blocking for form submission (disable inputs)
Success	Toast notification — bottom of screen on desktop, TOP of screen on the Technician PWA (to avoid competing with the primary action button and undo chip). Auto-dismiss after 3-4s for success; warning toasts may persist until dismissed	Non-blocking; page updates with new data
Error (client-side)	Inline field error below input; no toast	Fires on blur; does not block other fields
Error (server-side)	Field-level error OR generic error toast	Modal stays open; form stays available for correction
Empty state	Illustration + contextual headline + body copy + optional CTA	Each empty state is context-specific. Required empty states: "No assigned jobs — all caught up!", "No jobs pending schedule — all caught up!", "You're all caught up" (notifications), analytics-with-no-data, filtered-list-with-zero-results (distinct copy from genuine empty list), notification panel after "mark all read", dealer first-login job history
Real-time update	Status chip re-renders WITH a 100ms pulse (anim-instant); notification badge increments	Via WebSocket or polling; no full page reload required
WebSocket disconnect (desktop portal)	Subtle live indicator in nav (green dot = connected, grey = reconnecting). Non-blocking banner: "Live updates paused. Reconnecting..."	Auto-retry. On reconnect, silently refresh the current view and dismiss the banner. Office staff must never act on stale data without a visible signal

4.2 Undo System

Applies to Technician transitions only. 60-second window. See Undo Chip component for full details. The chip is a global sticky overlay anchored to the app shell — visible on every PWA screen during the window, never blocking swipe interactions on cards behind it.

Scenario	Behaviour
Transition not yet synced	Undo removes local queue entry; no server call required
Transition already synced	Undo sends rollback request to server
Rollback of completed/resolved	Payment row deleted atomically in same transaction
Rollback of needs_revisit	Revisit row DELETED; amber alert cleared
60 seconds expired	Chip dismisses; transition permanent; no undo available
Staff rollback (separate mechanism)	Modal with mandatory reason (≥ 10 chars); NOT the 60s chip; revisit row PRESERVED as historical record (resolves OQ-03)

4.3 Offline Sync System (Technician PWA)

Scenario	UI Behaviour
Network unavailable	Persistent 'Offline' banner pinned to top of screen at all times. Banner is NOT user-dismissible while still offline — it remains until connectivity is restored (resolves OQ-06)
Pending transitions	Sync queue indicator: floating pill showing "🔄 N transitions pending sync". Tapping the pill expands a detail sheet listing each queued action with its job reference and timestamp
Sync in progress	Indicator animates / updates (RefreshCw Lucide icon)
Sync success	Job state refreshed from server; queue indicator dismissed
Server rejects queued transition	Error toast with specific rejection reason; item stays in queue for manual resolution. Sync queue indicator changes to a RED error state — never silently decrements the count
App restart while offline	Queue is persistent (IndexedDB or service worker cache); not lost on restart
Payment + status queued together	Atomic single queue entry; both committed or both fail

4.4 Conflict Handling (HTTP 409)

Both jobs and payments use version counters for optimistic locking.

Scenario	UI Behaviour
409 received on any job/payment update	Show non-blocking toast: "This record was updated by someone else. Refreshing..."

Scenario	UI Behaviour
Auto-refresh	Fetch latest version from server; re-render the screen with updated data
User had pending changes in-flight	Open a bottom sheet (NOT a modal) showing "Your changes" vs "Current version" in a two-column diff view, with three options: Keep mine / Use latest / Review both. Do not silently discard the user's changes
Duplicate payment 409 (technician retry)	Treat as success — payment already recorded; do not show error

4.5 Notification Behaviour

Behaviour	Rule
Delivery	In-app only. No push notifications
Badge	Unread count on nav notification icon; hidden (not '0') when count = 0
Read state	Clicking marks <code>is_read = TRUE</code> , <code>read_at = NOW()</code> ; navigates to related job
Grouping	Notifications related to the same <code>job_id</code> (count ≥ 2) render as a single collapsible item ("3 updates on Job #AB123412"). Expanded items behave as individual notifications on tap
Filtering	Panel header chip row: All Unread Cancellations Assignments. Selection persists in URL query parameter
Pagination	Virtual scroll or paginate at 50 items
Deduplication	Backend only — 4 partial indexes; UI must not implement its own dedup
Role scoping	Each role receives only their targeted notification events
Dealer scoping	Dealer sees only <code>cancellation_request_outcome</code> events for their jobs
Urgent styling	<code>low_rating_received</code> , <code>pre_chronic_warning</code> , <code>chronic_flagged</code> , <code>needs_revisit_raised</code> render with their distinct accent colour and icon

4.6 Filter & View State Persistence (NEW)

All job-list surfaces (SH-02 and any KPI deep-linked filtered list) persist filter state in URL query parameters (e.g. `?status=needs_revisit&brand=X`). This enables shareable filtered views, browser back-

button support, and bookmark-ability — all at zero backend cost. The same approach applies to the notification panel filter chip selection.

4.7 Confirmation Patterns (NEW)

- Single-tap confirmations are reserved for low-risk actions (e.g. mark-all-read).
- High-risk technician actions (Complete Job, Mark Resolved) require a one-screen confirmation summary on TC-03 before final submit. See Pattern: Payment Collection.
- High-stakes administrative actions (status override, void/retain, reopen cancelled, direct cancellation) use modals with mandatory reasons (≥ 10 chars) and may not be dismissed without explicit choice when classed as Blocking.

5. Role-Based Rendering Rules

All routes are protected by JWT role guards. The following rules govern what is rendered per role.

5.1 Visibility vs Restriction

Rule

Unauthorised actions must NEVER appear as disabled buttons. They must be omitted entirely from the DOM. Disabled versions of unauthorised actions are prohibited.

5.2 Capability-to-Render Matrix

Capability	Owner	Office Staff	Technician	Dealer
Create jobs	✓ Render	✓ Render	— Omit	✓ Render (scoped)
Assign technicians	✓ Render	✓ Render	— Omit	— Omit
Batch schedule (OS-04)	✓ Render	✓ Render	— Omit	— Omit
Advance job status	✓ Render	✓ Render	✓ Render	— Omit
One-state rollback	✓ Render	✓ Render	— Omit	— Omit
Full status override	✓ Render	— Omit	— Omit	— Omit
Cancel job directly	✓ Render	✓ Render	— Omit	— Omit
Submit cancel request	— Omit	— Omit	— Omit	✓ Render
Approve cancel request	✓ Render	✓ Render	— Omit	— Omit
Record payment	✓ Render	✓ Render	✓ Render	— Omit
Edit payment method	✓ Render	✓ Render	— Omit	— Omit
Edit payment amount	✓ Render	— Omit	— Omit	— Omit
Void / retain payment	✓ Render	— Omit	— Omit	— Omit
Reopen cancelled job	✓ Render	— Omit	— Omit	— Omit
Manage brands/methods	✓ Render	— Omit	— Omit	— Omit
View analytics	✓ Render	✓ Render	— Omit	— Omit
Dealer analytics tab	✓ Render	✓ Render	— Omit	— Omit

Capability	Owner	Office Staff	Technician	Dealer
See VCID / customer identity	✓ Render	✓ Render	— Omit	— Omit
See technician name	✓ Render	✓ Render	— Omit	— Omit
See payment amount	✓ Render	✓ Render	✓ Render (own submission)	— Omit
See internal timeline events	✓ Render	✓ Render	— Omit	— Omit
See analytics data	✓ Render	✓ Render	— Omit	— Omit
avg_star_rating column	Conditional (review_mode ≠ off)	Conditional (review_mode ≠ off)	— Omit	— Omit
Customer profile screen (job history per VCID)	✓ Render	✓ Render	— Omit	— Omit
Command palette (Cmd+K)	✓ Render	✓ Render	— Omit	— Omit
Today's Focus toggle (dashboard)	✓ Render	✓ Render	— Omit	— Omit

Dealer analytics tab — resolves OQ-09

The Dealer analytics tab in OW-03 / OS-09 is visible to BOTH Owner and Office Staff. Office Staff have full analytics access aligned with their operational responsibilities.

5.3 Dealer Portal Isolation

- Dealer authenticates via a separate portal URL and dealer_credentials table — NOT the main users table.
- Dealer sees only jobs they submitted (scoped query by dealer_id).
- Dealer status labels are simplified (see Section 2: Status Chip component — Dealer-mapped variant).
- Dealer never sees: VCID, technician names, payment amounts, internal job_timeline events, or analytics data.
- Brand dropdown scoped to dealer_brands — no other brands visible.
- Dealer phone-lookup never exposes name, address, or confidence detail. "New Customer" / "Existing Customer" is the entire surface (resolves OQ-04).

- Dealer detail view (DL-03) shows scheduled_at and a simplified progress stepper (Submitted → Scheduled → In Progress → Completed) with the current stage highlighted. The label "In Progress" can be refined to "Technician en route" or "Work in progress" based on underlying in_transit vs in_process state, without exposing the technician's identity.
- Dealer KPI row (DL-01): Total Submitted (this month) | Active | Completed | Pending Cancellation.

5.4 Technician Portal Rules

- Technician sees only their assigned jobs.
- Punctuality derivation (visit_outcome, late_by_minutes) is computed by backend on 'I've Arrived' tap — UI does NOT display the outcome to the technician.
- Technician has no access to analytics, other technicians' jobs, customer VCIDs, or payment amounts after submission.
- is_chronic flag triggers owner/staff notification — technician receives no specific indication.
- Tab labels include badge counts: "Today (3) | Upcoming (1) | All (8)"; Today tab is highlighted when there are jobs not yet acknowledged.
- Job list (TC-01) supports swipe-right (next status action) and swipe-left (Call Customer) on each card.
- Job Action Screen (TC-02) wraps the address in a one-tap Navigate deeplink (Lucide MapPin) opening the device's native maps application.

6. Design-to-Code Mapping

6.1 Database Field → UI Field Mapping

DB Column (jobs table)	UI Label	Display Rules
id	Job ID / Reference	Shortened to first 8 chars in lists; full UUID in detail header with Copy icon. Copy action shows a 1.5s "Copied!" tooltip
type	Job Type	'Installation' or 'Complaint' chip; uses color-type-installation or color-type-complaint
source	Source	'Direct' or 'Via Dealer' chip; uses color-source-direct or color-source-dealer; show dealer business_name if via_dealer
status	Status	Status chip per icon + colour + label convention; real-time update via WebSocket/poll with 100ms pulse
brand_id	Brand	Brand name + colour_hex swatch circle; falls back to color-brand-fallback (#9E9E9E) when colour_hex is NULL — resolves OQ-10
dealer_id	Dealer	business_name shown to staff/owner; hidden from dealer/technician view
vcid	Customer Identity	Shown in owner/staff view only; hidden from dealer and technician; clickable link to customer profile screen
customer_name	Customer Name	Editable by owner/staff
address	Address	On Technician PWA, wrapped in a Navigate deeplink to the device maps application
is_repeat	Repeat tag	Boolean → show 'Repeat' tag if TRUE; tooltip shows window used
is_frequent	Frequent tag	Boolean → 'Frequent' tag if TRUE
is_chronic	Chronic tag	Boolean → 'CHRONIC' tag in red if TRUE
scheduled_at	Scheduled	Human-readable date+time in local timezone. Within-2-hour same-day jobs show amber relative countdown; overdue >30 min show red elapsed
actual_arrival	Arrived At	[SYSTEM] Shown in timeline and visit outcome panel

DB Column (jobs table)	UI Label	Display Rules
visit_outcome	Visit Outcome	on_time late no_show rescheduled — shown in analytics/detail for owner/staff only
late_by_minutes	Late By	Formatted: '23 minutes late' — analytics only
cancellation_reason	Cancellation Reason	Required for all cancellations (≥10 chars); displayed in job detail history
version (hidden)	—	Used internally for optimistic locking — never exposed in UI

6.2 Backend-Computed Fields ([SYSTEM])

These values are written by the backend. The UI displays them only and must not provide edit affordance for them.

Field	Computed By	UI Treatment
vcid	Backend on customer resolution	Display with link to customer profile; never editable
is_repeat, is_frequent, is_chronic	Backend evaluation against system_config thresholds	Display as tags; not editable; point-in-time snapshots
visit_outcome	Backend on in_process tap (actual_arrival vs scheduled_at + grace)	Shown in analytics/detail for owner/staff only; hidden from technician
late_by_minutes	Backend on late arrival	Analytics display only
collected_at	trg_set_collected_at trigger on INSERT	Timeline display; no UI input
is_low_rated	Generated column (rating ≤ 2)	Auto-renders low-rating flag on job card/detail header; triggers low_rating_received notification with urgent styling; no manual toggle
read_at	Backend on notification read	Used for UI state only; not displayed
pre_chronic_warning fired	Backend on revisit_sequence = 2	Triggers pre_chronic_warning notification to Owner with urgent styling

6.3 system_config → UI Behaviour

Config Key	UI Effect
repeat_complaint_window_days	Window used to determine is_repeat flag; shown in tooltip on Repeat tag
frequent_complaint_threshold	Threshold for is_frequent flag; referenced in analytics labels
frequent_complaint_window_days	Window for is_frequent evaluation
punctuality_grace_period_mins	Used in visit_outcome derivation; not shown to technician
customer_review_mode	off: review tab hidden, avg_star_rating hidden, no review sheet after completion. optional: review sheet with Skip button (tap-outside and back also count as Skip). mandatory: review sheet without Skip; tap-outside and back are blocked; 'Review Pending' indicator shown until submission or 48h expiry
standard_job_duration_mins	Scheduling conflict check window; used in conflict detection query and Batch Schedule auto-spacing preview
pending_schedule_amber_days_installation (NEW)	Soft amber threshold for Days Waiting on installation rows. Default: 2
pending_schedule_red_days_installation (NEW)	Hard red threshold for Days Waiting on installation rows. Default: 4
pending_schedule_amber_days_complaint (NEW)	Soft amber threshold for Days Waiting on complaint rows. Default: 1
pending_schedule_red_days_complaint (NEW)	Hard red threshold for Days Waiting on complaint rows. Default: 2
cancellation_request_escalation_minutes (NEW)	Threshold after which an unactioned cancellation_requested job escalates to a persistent dashboard banner. Default: 30

6.4 WebSocket / Polling Events → UI Re-renders

Event	UI Response
Job status changed	Status chip re-renders on Job Detail and any open job list rows, with 100ms pulse (anim-instant)
Cancellation request submitted (dealer)	Cancellation Request banner appears on job action panel; technician buttons frozen; technician's job card flips to frozen state with the defined message

Event	UI Response
Cancellation request decided (staff/owner)	Dealer portal updates job status; dealer notification created
New notification	Nav badge increments; panel list prepends new item (or appends to existing group when same job_id)
Job auto-rejected (stale cancel request)	Cancellation Request banner removed; action panel restored; dealer notified. system_event of type cancellation_auto_rejected appears in timeline (resolves OQ-05)
WebSocket disconnect	Live indicator flips to grey; non-blocking banner: "Live updates paused. Reconnecting..."
WebSocket reconnect	Live indicator returns to green; current view silently refreshes; banner dismissed

7. Navigation Structure

This section defines the navigation pattern for each role's surface. The original v1.0 spec defined layout archetypes per role but omitted navigation explicitly — these patterns are now binding.

Role	Navigation Pattern
Owner / Office Staff	Persistent left sidebar on lg+ breakpoints; collapses to top nav on md. "Log New Job" persistent top-right primary button. Command palette (Cmd+K / Ctrl+K) globally available — searches across jobs, screen navigation, and common actions (Log New Job, Open Pending Queue). Zero-navigation-depth pattern appropriate for power users managing hundreds of jobs
Technician PWA	Bottom tab bar: Today Jobs All Jobs Notifications. Tabs include badge counts ("Today (3)") and Today is highlighted while jobs remain unacknowledged. Bottom nav minimises thumb travel on xs screens. The global undo chip lives at the app shell level — see Section 2
Dealer	Minimal top nav: Jobs Submit Notifications. No sidebar — narrow scope of dealer actions does not warrant one

7.1 Command Palette (Owner / Office Staff Desktop)

- Trigger: Cmd+K (macOS) / Ctrl+K (Windows, Linux).
- Categories: Jobs (search by ID, customer name, brand), Screens (Dashboard, Pending Schedule Queue, Analytics, Brand Management, etc.), Actions (Log New Job, Open Notifications, Mark All Read).
- Keyboard navigation: Up/Down arrows to traverse, Enter to select, Esc to dismiss.
- Renders as a centred overlay with a search input, grouped results, and recent items at the top of the empty state.

7.2 Today's Focus Toggle (Owner / Office Staff Dashboard)

The OW-01 layout places KPI cards, active jobs table, amber alert panel, and analytics summary on one page. Today's Focus is a toggle (top right of the dashboard) that collapses the full job table and surfaces only the operationally urgent view: KPI row + Amber Alert Panel + Pending Schedule queue. The full table remains the management view; the toggle preference persists per user.

8. Role-Based Improvements & Workflow Optimisations

These requirements consolidate role-specific surface treatments and behavioural improvements that span more than one component or pattern. Cross-references to the primary component or pattern definition appear inline.

8.1 Owner & Office Staff

Job Detail Action Panel — Visual Hierarchy

The right-side action panel groups workflow actions (Advance Status, Roll Back) alongside administrative actions (Assign Technician, Cancel Job) and payment actions. These are visually grouped into three labelled sections separated by a subtle divider:

- Workflow Actions — Advance Status, Roll Back Status, Full Status Override (Owner)
- Administrative — Assign Technician, Reassign Technician, Cancel Job, Reopen Cancelled Job (Owner), View Timeline
- Payment — Edit Payment Method, Edit Payment Amount (Owner), Void / Retain (Owner)

Grouping prevents accidental cancellations when reaching for the rollback button.

Customer Profile Screen (NEW required surface)

v1.0 referenced a "link to customer profile view" for VCID without defining the screen. The minimal customer profile screen is now a required surface (Owner and Office Staff only):

- Header: VCID, primary phone, all linked phone numbers, customer_name (current), address (most recent)
- Job history: list of all jobs for this VCID, newest-first, with status chip, scheduled_at, brand, technician name
- Indicators: is_repeat / is_frequent / is_chronic tags shown at the header level
- Read-only — no edit affordance on the profile itself; edits happen from individual job records

Analytics Export

- CSV export button per analytics tab, positioned at the top right as a secondary action.
- Export scope: current date range and any applied filters.
- File naming: cooldesk-[tab-name]-[date-range].csv.

Loading Stagger

On dashboard load, KPI cards render in a defined order to surface actionable information sooner:

- 46. Amber Alerts and Total Active Jobs (load first)
 - 47. Today's Revenue and Completion Rate (load second)
 - 48. Remaining KPIs
-

8.2 Technician PWA

Job List Tabs

- Today / Upcoming / All tabs include unread/active counts: "Today (3) | Upcoming (1) | All (8)". Today is highlighted when unacknowledged jobs exist.
- Counts come from the same data the list already fetches — render-only addition.

Swipe Actions on Job Cards

- Swipe-right on a non-frozen card triggers the next logical status action (same as primary action button).
- Swipe-left triggers Call Customer (opens device dialer).
- On frozen cards, swipe is disabled; only the call-office affordance remains active.

Navigate Deeplink

- Address on TC-02 wraps a Lucide MapPin Navigate button that opens the device's native maps application via the appropriate URL scheme (<https://maps.google.com/?q={address}> on Android, geo: scheme honored on supported devices, Apple Maps on iOS).
- This is the single most-used action before acknowledging a job.

Touch-Friendly Reason Selection (TC-04 Needs Revisit)

- Replace radio groups with full-width tappable chip/card options for each reason variant.
- Minimum 48px tappable height (WCAG 2.5.5).
- Clear selected vs unselected state via fill colour and border, not just a small radio dot.

Route Optimisation (Today Tab)

- Optional "Optimise Route" button on the Today tab reorders jobs by geographic proximity using the Haversine formula against stored addresses.
- Computed client-side; shows estimated drive time between stops; user can revert to scheduled order with one tap.
- No backend change required.

Sync Queue Detail

- Floating pill: "🔄 N transitions pending sync". Tapping expands a detail sheet listing each queued action with job reference and timestamp.
- Failed queued transitions (server-rejected after reconnect) flip the pill to a red error state. The count is never silently decremented.

8.3 Dealer

DL-01 — Minimal KPI Row

Above the job history table, render a minimal at-a-glance KPI row:

- Total Submitted (this month)
- Active
- Completed
- Pending Cancellation

These are dealer-scoped counts requiring no sensitive data exposure and significantly reduce "how many jobs are active?" enquiries to the office.

DL-02 — Post-Submission Navigation

- On successful submission, show: reference ID + primary CTA "View Job" (navigates to DL-03 for the new job) + secondary CTA "Submit Another".
- The success state is never a dead end.

DL-03 — Simplified Progress Stepper

Detail view shows a horizontal progress stepper:

Submitted → Scheduled → In Progress → Completed

- Current stage highlighted in color-primary-600.
- "In Progress" can be refined to "Technician en route" or "Work in progress" based on underlying `in_transit` vs `in_process` — without exposing technician identity.
- `scheduled_at` is shown in the detail (non-sensitive). Dramatically reduces "where is my job?" inbound queries.

DL-04 — Cancellation Request Submission Confirmation

- After submitting a cancellation request, show in-page success state: "Your cancellation request has been submitted. You'll be notified once a decision is made."
- Includes the job reference.
- Currently the dealer only sees the status change in the list with no in-context acknowledgment — this gap is closed.

8.4 Behavioural Design

Non-Linear Undo Chip Progress

The undo chip's visual progress ring decrements non-linearly: slow for the first 40 seconds, accelerating in the final 20. Behavioural research suggests users are more likely to use an undo option when the window appears larger than it is. This increases utilisation for genuine mistakes without extending the actual 60-second wall-clock window. See Component: 60-Second Undo Chip.

Cancellation Request Urgency Escalation

If a `cancellation_requested` job is not actioned within 30 minutes (configurable via `cancellation_request_escalation_minutes` in `system_config`; default 30), the request is escalated to a persistent amber banner on the dashboard with inline Approve/Reject CTAs. Prevents cancellation requests from being buried in notification noise during busy periods. See Component: Amber Alert Panel.

Pre-Chronic Warning

The `is_chronic` flag triggers at the 3rd revisit. By that point the situation has already escalated. A `pre_chronic_warning` notification fires at the 2nd revisit (`revisit_sequence = 2`): "This job has had 2 revisits. One more will flag it as Chronic." This gives owners an actionable window to intervene before the threshold is crossed. See Component: Notification Centre.

Smart Customer Pre-Fill for Repeat VCIDs

In Step 3 of OS-02 Job Creation, when a known customer is matched in Step 2, the customer's most-recently-used brand is offered as a one-tap suggestion chip: "Last brand: [X] — use again?" Reduces form entry time for repeat customers without imposing a default. See Pattern: Job Creation Flow.

Smart Technician Suggestion for Repeat Complaints

When a complaint is submitted for a customer with a prior completed job, the technician picker pre-suggests the original technician at the top of the list: "Assign to [Name] — completed prior installation

for this customer." Familiarity reduces complaint resolution time. See Pattern: Technician Assignment & Scheduling Conflict Check.

8.5 Microcopy & Empty States

Microcopy revisions adopted in v2.0:

Surface	Original v1.0 copy	v2.0 copy
Job type mismatch error	"Job type mismatch. Please reload and try again."	"Something went wrong on our end. Your form data is saved — please try again."
Void-or-Retain options	(no descriptions)	Void: "Remove this payment record entirely" / Retain: "Keep the payment record without reversing it"
Empty pending schedule	"No jobs pending schedule"	"No jobs pending schedule — all caught up!" with CheckCircle icon
Network failure on dealer submit	(generic)	"Your job couldn't be submitted due to a connection issue. Your form data has been saved. Tap Retry to try again." Show a retry counter on repeated attempts
Inactive account login	"Your account is inactive. Please contact support."	Unchanged — appropriately vague, does not confirm account existence

Empty states required (consistent pattern: contextual illustration + headline + body copy + optional CTA):

- Analytics with no data in the selected date range
- Job list with active filters returning zero results (distinct copy from genuinely empty list)
- Notification panel after bulk "mark all read" action
- Dealer job history on first login (before any jobs submitted)
- Technician "No assigned jobs — all caught up!" with CheckCircle illustration
- Pending Schedule Queue when count = 0

9. Resolved Decisions Log

Section 7 of v1.0 listed ten open questions and four missing design system elements. All are resolved below as binding requirements. This section is retained as a traceable record of how each was decided.

9.1 Resolved Open Questions

Ref	Area	Decision
OQ-01	Amber Alert KPI zero state	Show green zero-state with checkmark and "No revisits pending" label. The card is never hidden — its position is a navigational anchor and hiding it creates layout shift. Implemented in Component: KPI Card and Component: Amber Alert Panel
OQ-02	Pending Schedule Days Waiting threshold	Two-tier configurable threshold: amber/red at 2/4 days for installation, 1/2 days for complaint. Exposed in OW-04 (System Config) as four new keys (see §6.3). Implemented in Pattern: Technician Assignment & Scheduling Conflict Check
OQ-03	Staff rollback from needs_revisit	Revisit row is PRESERVED as historical record on staff rollback. The 60-second technician undo deletes the revisit row; staff rollback does not. Two distinct mechanisms — technician undo is reversible mistake-correction; staff rollback is an audited administrative action that should leave a trail
OQ-04	Dealer visibility on VCID conflicts	No additional disclosure beyond the existing "New Customer / Existing Customer" indicator. VCID conflict resolution is internal staff work; further disclosure to dealers creates support burden without giving them actionable information
OQ-05	Auto-rejection trigger mechanism	Service-layer check on every status transition — NOT a background job. Atomic with the causing transition, eliminating the inconsistency window. system_event type = cancellation_auto_rejected, actor = System, reason = "Job lifecycle advanced beyond cancellation request". Implemented in Pattern: Cancellation Flow
OQ-06	Offline banner dismissal	Pinned, not user-dismissible while still offline. The banner remains until connectivity is restored. Dismissibility while disconnected creates risk of stale-state errors
OQ-07	Withdrawal reason minimum character count	10-character minimum enforced — consistent with all other reason textareas (cancellation, void, rollback). A consistent floor across reason fields keeps audit records meaningful and the validation rule easy to reason about
OQ-08	Optional review sheet dismissal	Tap-outside and device back both count as Skip in optional mode. In mandatory mode, tap-outside and back are blocked; technician must generate the link

Ref	Area	Decision
OQ-09	Dealer analytics tab access	Visible to BOTH Owner and Office Staff. Office Staff have full analytics access aligned with their operational responsibilities. Implemented in §5.2
OQ-10	Brand colour swatch fallback	Neutral grey (#9E9E9E) via new color-brand-fallback token. Applied uniformly wherever a brand swatch is rendered when colour_hex is NULL

9.2 Resolved Design System Gaps

Element	Decision
Typeface / font family	Inter (or equivalent system sans-serif). Fallback stack: Inter, -apple-system, "Segoe UI", Roboto, sans-serif. See §1.2
Icon library	Lucide Icons system-wide. Status group → icon assignments defined in §1.5
Animation / transition timing	Three tokens: anim-instant (100ms), anim-standard (250ms), anim-countdown (1000ms). See §1.6
Brand colour swatch fallback	color-brand-fallback (#9E9E9E). See §1.1 and §6.1
Days Waiting amber threshold	Configurable per job type via four new system_config keys. See §6.3
Toast duration and positioning	Bottom on desktop; top on Technician PWA (avoids competing with the primary action button and undo chip). 3-4s auto-dismiss for success; warning toasts may persist until dismissed

9.3 Critical UX Issues Resolved in v2.0

The following issues identified in the UX review are now first-class requirements. Each is listed with its primary location in this document.

Issue	Where addressed in v2.0
Status chip colour-only differentiation (WCAG 1.4.1)	§1.5 Icon System; Component: Status Chip
Frozen job card has no explanation to the technician	Component: Job Card (frozen state, message, call-office affordance)
No draft / auto-save on OS-02 Job Creation	Pattern: Job Creation Flow

Issue	Where addressed in v2.0
Undo chip cross-screen persistence unspecified	Component: 60-Second Undo Chip (global sticky overlay anchored to app shell)
Blocking vs dismissible modals — no visual distinction	Component: Modal / Confirmation Dialog (Distinguishing Blocking from Dismissible Modals table)
Amber Alert Panel — no sort order defined	Component: Amber Alert Panel (Default Sort Order)
Payment Form — no confirmation step	Pattern: Payment Collection (one-screen confirmation summary)
WebSocket disconnect not defined for desktop portal	§4.1 (WebSocket disconnect row)
Filter state not persisted across navigation	§4.6 (Filter & View State Persistence)
Notification panel — no grouping or pagination	Component: Notification Centre; §4.5
Dealer portal — no ETA or progress context	§5.3; §8.3 (DL-03 progress stepper)
VCID phone lookup — confidence signalling insufficient	Component: Form Field System (Phone Lookup Behaviour)
Toast placement conflict on Technician PWA	§4.1 (Success row notes top placement on PWA)
409 conflict diff prompt unspecified	§4.4 (bottom sheet, 3-way choice)
Touch targets on PWA below WCAG 2.5.5	§1.3 (space-12 = 48px minimum); §8.2 (touch-friendly reasons)
iOS safe-area-inset-bottom for primary action button	§1.3 callout; Component: Primary Action Button
Colour token collision (in_process / revisit both #F57F17)	§1.1 (color-status-revisit re-keyed to #EF6C00)
color-primary-100 dual purpose (structural + semantic info)	§1.1 (new color-surface-info token)
Job Type / Source chips have no defined tokens	§1.1 (color-type-installation, color-type-complaint, color-source-direct, color-source-dealer)
color-status-pending shared between pending_schedule and new	§1.1 (color-status-new split for independent evolution)
Star rating component dimensions undefined	Component: Form Field System (Star selector — 48px, 8px gap, defined fill/press states)
KPI card trend indicators optional but operationally important	Component: KPI Card (trends MANDATORY for Today's Revenue and Completion Rate)

Issue	Where addressed in v2.0
Type scale compresses h3 and body-lg to identical sizes	§1.2 (type-h3 raised to 14pt)
type-label illegible on xs in outdoor conditions	§1.2 (12pt mobile minimum); Component: Status Chip
type-caption below WCAG 1.4.4 minimum at 10pt	§1.2 (raised to 11pt system floor)